

EC400 – Static Optimization and Fixed Points Syllabus September 2026

This part of course is based on three live lectures (2.5 hours each) and on three classes (2.5 hours each). Bespoke recordings are available on Moodle. Lectures revise essential mathematical notions for static optimization and fixed point theorems. Classes show how to apply these notions to solve relevant problems.

Course Outline

Lecture 1: Tools for Optimization

- Quadratic Forms
- Determinant
- Taylor's Expansion
- Concavity and Convexity
- Quasi-Concavity and Quasi-Convexity

Lecture 2: Optimization

- Unconstrained Optimization
- Constrained Optimization: Equality Constraints
- Lagrange Theorem
- Constrained Optimization: Inequality Constraints
- Kuhn-Tucker Theorem
- Arrow-Enthoven Theorem

Lecture 3: Comparative Statics and Fixed Points

- Envelope Theorem
- Implicit Function Theorem
- Correspondences
- Fixed Point Theorems
- Theorem of the Maximum

Suggested Textbooks

- Velleman, *How to Prove It*, 3rd edition, Cambridge University Press, 2019.
- Strang, *Introduction to Linear Algebra*, 6th edition, Wellesley-Cambridge Press, 2023.
- Simon & Blume, *Mathematics for Economists*, Norton, 1994.
- Takayama, *Mathematical Economics*, 2nd edition, Cambridge University Press, 1985.
- Ok, *Real Analysis with Economic Applications*, Princeton University Press, 2007.
- Chiang, *Elements of Dynamic Optimization*, McGraw-Hill, 1992.
- Acemoglu, *Introduction to Modern Economic Growth*, Princeton University Press, 2009.

Exercises

- Three problem sets to test background and preparation.
- Access to past exam paper to prepare for the exam.